INTERIM FEDERAL SPECIFICATION
FILES, ROTARY

This Interim Federal Specification was developed by the Standardization Division, Federal Supply Service, General Services Administration, Washington, D. C. 20406, based upon currently available technical information. It is recommended that Federal agencies use it in procurement and forward recommendations for changes to the preparing activity at the address shown above.

1. SCOPE AND CLASSIFICATION

1.1 Scope. - This specification covers rotary files used with portable and stationary power operated rotating equipment and are generally used for abrading and smoothing metals and other materials.

1.1.1 Federal specification coverage. - Federal specifications do not include all varieties of the commodity as indicated by the title of the specification, or which are commercially available but are intended to cover only those generally used by the Federal Government.

1.2 Classification.

1.2.1 Types. - Rotary files shall be of the following types, as specified (see 6.2).

Type I. - Tree, pointed end.
Type II. - Tree, radius end.
Type III. - Cylinderical, ball end.
Type IV. - Cylinderical, flat end.
Type V. - Cone.
Type VI. - Ball.
Type VII. - Barrel.
Type VIII. - Inverted cone.

2. APPLICABLE SPECIFICATIONS, STANDARDS, AND OTHER PUBLICATIONS.

2.1 Specifications and standards. - The following specifications and standards, of the issues in effect on date of invitation for bids, form a part of this specification:

Federal Specifications:
- QQ-T-590 - Tool Steel, High Speed.
- PPP-B-566 - Boxes, Folding, Paperboard.
- PPP-B-585 - Boxes, Wood, Wirebound.
- PPP-B-591 - Boxes, Fiberboard, Wood-Cleated.
- PPP-B-601 - Boxes, Wood, Cleated-Plywood.
- PPP-B-621 - Boxes, Wood, Nailed and Lock-Corner.
- PPP-B-636 - Box, Fiberboard.
- PPP-B-640 - Boxes, Boxes, Fiberboard, Corrugated, Triple, Wall.
- PPP-B-665 - Boxes, Paperboard, metal stayed.
- PPP-B-676 - Boxes, Set-up, Paperboard.

Federal Standards:
- Fed. Std. No. 102 - Preservation Packaging, and Packing Levels.
- Fed. Std. No. 123 - Marking for Domestic Shipment (Civilian Agencies).
- Fed. Std. No. 151 - Metals, Test Methods.

(Activities outside the Federal Government may obtain copies of Federal Specifications, Standards, and Handbooks as outlined under General Information in the Index of Federal Specifications and Standards and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U.S. Government Printing Office, Washington, D.C., 20402.

(Single copies of this specification and other product specifications required by activities outside the Federal Government for bidding purposes are available without charge at the General Services Administration Regional Offices in Boston, New York, Washington, D.C., Atlanta, Chicago, Kansas City, Mo., Dallas, Denver, San Francisco, Los Angeles, and Seattle, Wash.

(Federal Government activities may obtain copies of Federal Specifications, Standards, and Handbooks and the Index of Federal Specifications and Standards from established distribution points in their agencies.)

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Military Specifications:
MIL-P-116 - Preservation, Methods of

Military Standards:
MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129 - Marking for Shipment and Storage.

(Copies of Military Specifications and Standards required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS.

3.1 Illustrations. The illustrations shown herein are for the convenience of identification and are not intended to preclude the purchase of files which are otherwise in accordance with the requirements of this specification.

3.2 Shape and Dimensions. - The files shall conform to the configurations shown in figures 1 through 8 with tolerances for dimensions as follows: diameter of head is plus or minus .010 inch; length of head is plus or minus 1/32 inch, and over-all length is plus or minus 1/4 inch. All dimensions are in inches.

3.3 Material. - The files shall be made of high speed tool steel conforming to QQ-T-590, Class M2.

3.4 Hardness. - The files shall be hardened to not less than 62 and not more than 65 on the Rockwell "C" scale, (see 4.4.1).

3.5 Teeth. - The teeth of the files shall be machine cut with right hand spiral and ground after hardening. The number of teeth per inch shall be 16, plus or minus 2 teeth, and shall be counted at the largest diameter of the cutting head at right angles to the helix.

3.6 Shank. - The shank diameter shall be 0.2500 inch, plus .0000, minus .0008 inch and the end may be either chamfered, with radius or square.

3.7 Concentricity. - The shank diameter shall be concentric with the cutting head diameter within 0.010 inch total indicator reading (see 4.4.3).

3.8 Performance. - The files shall be capable of withstanding the performance test as specified in 4.4.2.

3.9 Preservation. - Files shall have an anti-corrosive preservation finish applied to entire surfaces (see 5.1.1.2).

3.10 Workmanship. - Workmanship shall be first class in every respect. Files shall be free of rust, cracks, burrs and other imperfections which may impair their serviceability or durability.

3.11 Identification marking. - Each file shall be marked on the shank, in a plain and permanent manner with the name of the manufacturer, or with a trademark of known character so that the source of manufacture can be readily determined. Such markings may include the manufacturers identification part number.

4. SAMPLING, INSPECTION, AND TEST PROCEDURES.

4.1 Responsibility for inspection. - Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Sampling.

4.2.1 Inspection lot. - All files offered for acceptance at one time shall be considered a lot for purposes of acceptance inspection.

4.2.2 Sampling for visual and dimensional inspection. - A random sample of files shall be selected from each lot in accordance with MIL-STD-105 at inspection level II. The AQL shall be 1.5 percent defective.

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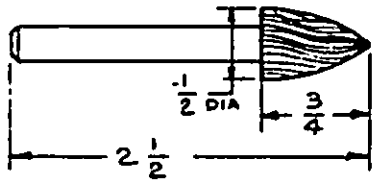


Figure 1. - Type I, Tree Pointed End

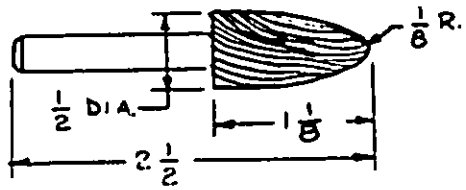


Figure 2. - Type II, Tree, Radius End

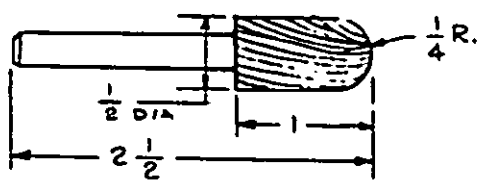


Figure 3. - Type III, Cylinderical, Ball End

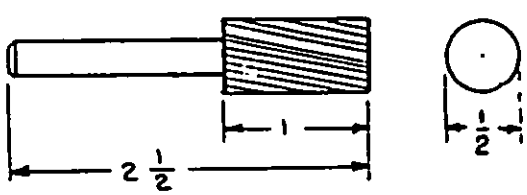


Figure 4. - Type IV, Cylinderical, Flat End

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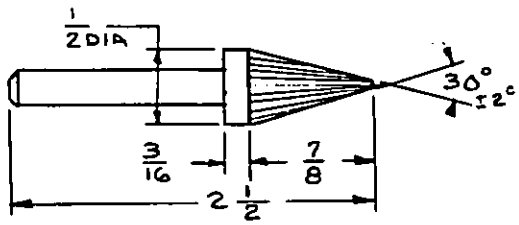


Figure 5, - Type V, Cone

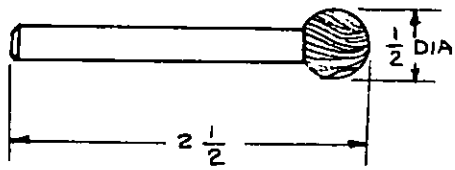


Figure 6, - Type VI, Ball

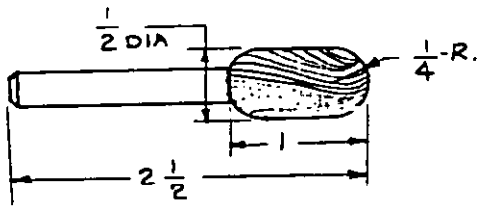


Figure 7. - Type VII, Barrel

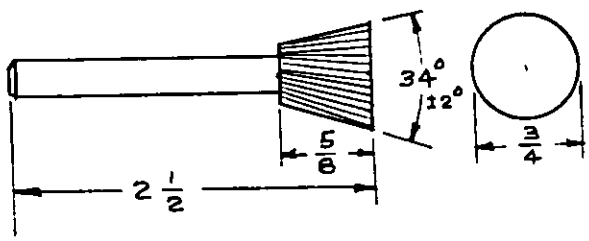


Figure 8. - Type VIII, Inverted Cone

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4.2.3 Sampling for acceptance testing. - A random sample of files shall be selected from each lot in accordance with MIL-STD-105 at inspection level S-4. The AQL shall be 2.5 percent defective.

4.3 Inspection and tests.

4.3.1 Visual and dimensional inspection. - Each sample file selected in accordance with 4.2.2 shall be visually and dimensionally inspected to determine conformance to the requirements stated in 3.2, 3.5, 3.6, 3.9, 3.10, and 3.11. Any file containing one or more defect shall be classified a reject.

4.3.2 Inspection test. - Each sample file selected in accordance with 4.2.2 and determined to conform to 4.3.1 shall be tested to determine conformance to the requirements of 3.4, 3.7, and 3.8. Any file that fails either test shall be considered a reject.

4.4 Test procedures.

4.4.1 Hardness test. - Hardness test shall be conducted in accordance with method 243 of Federal Test Method Standard No. 151.

4.4.2 Performance test.

4.4.2.1 Side of head. - Tests shall be made on a milling machine or similar device. The feed shall be that produced by a 5-pound weight operating a cord attached to a carriage mounted on ball bearings. The file shall rotate at 5700 revolutions per minute. The material to be cut shall be a piece of cold rolled steel, 0.250 inch thick, SAE 1020, with a Brinell hardness of 121. The end of the sheet shall be centered on the length of the head of the file. The axis of the file shall be perpendicular to the side of the plate and a series of eighteen cuts, one-half inch in length, shall be made. A recording device shall be used to time each of the eighteen cuts. The amount of time consumed in completing the eighteenth cut shall not exceed 2.5 times that required to complete the second cut. Any file that exceeds this value shall be considered defective.

4.4.3 Concentricity. - A comparator, Shadowgraph or similar instrument shall be used to conduct this test. The file shall be rotated slowly through one complete revolution. The difference in the maximum and the minimum reading shall not exceed 0.010 inch. Any file that exceeds this dimension shall be considered defective.

4.5 Inspection of preparation for delivery requirements. - An inspection shall be made to determine that the packaging, packing and marking comply with the requirements in section 5 of this specification. Defects shall be scored in accordance with table 1. The sample unit shall be one shipping container fully prepared for delivery. Sampling shall be in accordance with MIL-STD-105. The lot size shall be the number of shipping containers in the end item inspection lot. The inspection level shall be S-2 and the AQL shall be 4.0 defects per hundred units.

Table I Classification of preparation for delivery defects.

Examine	Defects.
Markings (exterior and interior).	Omitted; incorrect; illegible; improper size, location, sequence, or method of application.
Materials	Any component missing, or damaged.
Workmanship	Inadequate application of components such as incomplete closure of container flaps, loose strapping, inadequate stapling. Distortion of container.

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5. PREPARATION FOR DELIVERY
(For civil agency procurement the definitions and application of levels of packaging and packing shall be in accordance with Fed. Std. No. 102).

5.1 Packaging. - Packaging shall be level A, B or C as specified (see 6.2).

5.1.1 Level A.

5.1.1.1 Cleaning and drying. - The files shall be cleaned by process C-3 and dried by procedure D-1 in accordance with MIL-P-116.

5.1.1.2 Preservative application. - The files shall be coated with P-8 preservative in accordance with MIL-P-116.

5.1.1.3 Unit packaging. - Each file shall be packaged in accordance with method IC-1 of MIL-P-116.

5.1.1.4 Intermediate packaging. - Unless otherwise specified (see 6.2) 12 unit packages of files of like description shall be packaged in a box conforming to PPP-B-566, PPP-B-665, or PPP-B-676.

5.1.2 Level B.

5.1.2.1 Unit packaging. - Each file shall be packaged in a close-fitting box conforming to PPP-B-566, PPP-B-665, PPP-B-676, or in a plastic tube.

5.1.2.2 Intermediate packaging. - Twelve unit packages of files of like description shall be packaged in a close-fitting box conforming to PPP-B-566, PPP-B-665 or PPP-B-676.

5.1.3 Level C. - The files shall be packaged in accordance with the suppliers commercial practice.

5.2 Packing - Packing shall be level A, B or C as specified (see 6.2).

5.2.1 Level A. - Ten intermediate packages of files of like description shall be packed in a close-fitting box conforming to PPP-B-636, V3c. The box shall be closed, waterproofed sealed and strapped in accordance with the appendix to the box specification.

5.2.1.1 Additional packing. - When specified (see 6.2), the fiberboard boxes shall be overpacked in a close-fitting box conforming to PPP-B-585, class 3; PPP-B-591, class II; PPP-B-601, type III; PPP-B-621, class 2; or PPP-B-640, class 2, grade A. The box shall be closed and strapped in accordance with the applicable specification or appendix thereto. The gross weight shall not exceed 200 pounds. Strapping of the individual fiberboard boxes shall not be required when they are overpacked.

5.2.2 Level B. - Twelve intermediate packages of files of like description shall be packed in a close-fitting box conforming to PPP-B-636, class-domestic. The box shall be closed in accordance with the appendix to the box specification.

5.2.3 Level C. - The files shall be packed to insure carrier acceptance and safe delivery at destination in containers complying with the rules and regulations applicable to the mode of transportation.

5.3 Standard Pack (Civil Agency Procurements). - The Standard Pack for civil agency procurements shall be as specified in paragraphs 5.1.2 and 5.2.2 (see 6.3).

5.4 Marking.

5.4.1 Civil agencies. - In addition to markings required by the contract or order (see 6.2), the packages and shipping containers shall be marked in accordance with Fed. Std. No. 123.

5.4.2 Military activities. - In addition to markings required by the contract or order (see 6.2) the packages and shipping containers shall be marked in accordance with MIL-STD-129.

6. NOTES

6.1 Intended use. - The files described herein are used on ductile materials, particularly non-ferrous metals and some grades of plastic. These files are used to the best advantage at the speeds and with materials as follows: Mild steel- 2300 r.p.m.; cast iron - 3500 r.p.m; bronze - 7500 r.p.m.; aluminum - 10,000 r.p.m. and magnesium - 15,000 r.p.m.

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6.2 Ordering data. - Procurement documents should specify the following:

(a). Title, number, and date of this specification.
(b). Type of file required (see 1.2.1).
(c). Level of packaging and packing required (see 5.1 and 5.2).
(d). Intermediate packaging, if different (see 5.1.1.4).
(e). Additional packing if required (see 5.2.1.1).
(f). Special marking if required (see 5.4).

6.3 Standard pack for civil agencies. - The standard pack requirements in 5.3 are intended for use in procurements of stores stock replenishments. Procuring officers should use the standard pack requirements when it is known that the material will be shipped from a supplier to a domestic warehouse, supply depot or intermediate storage point for temporary storage, subsequent issue or shipment to eventual user.

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